

Introduction to AI Governance

WHY THIS MODULE MATTERS

AI is already operating inside most organisations, whether leadership has approved it or not. This module gives you the shared vocabulary and the lay of the land before anything else in this course will make sense.

1.1 Welcome to the Course

Artificial intelligence is no longer an experimental capability confined to research labs; it now shapes hiring decisions, credit approvals, medical triage, content moderation and public administration across the United Kingdom.

As AI systems take on more consequential roles, the gap between what a system can do and what it should do becomes the central question for anyone building, deploying or overseeing these technologies. This course exists to help close that gap.

This certification course treats AI ethics and governance as a practical discipline rather than an abstract debate. It is designed for practitioners — engineers, product managers, risk and compliance professionals, policy leads and executives working in or with UK organisations — who need working frameworks, not just vocabulary. Rather than working through abstract principles alone, this course returns repeatedly to a small number of real, well-documented cases where AI governance succeeded or failed in public view, and asks what specifically went right or wrong.

1.2 Course Structure and Study Time

The course totals approximately 29–35 hours of study, organised around a simple arc: why ethics matters in AI development, how systems fail when it is neglected, what governance looks like in practice, how UK data protection law applies, how to implement responsible AI in practice, and how to carry these lessons forward.

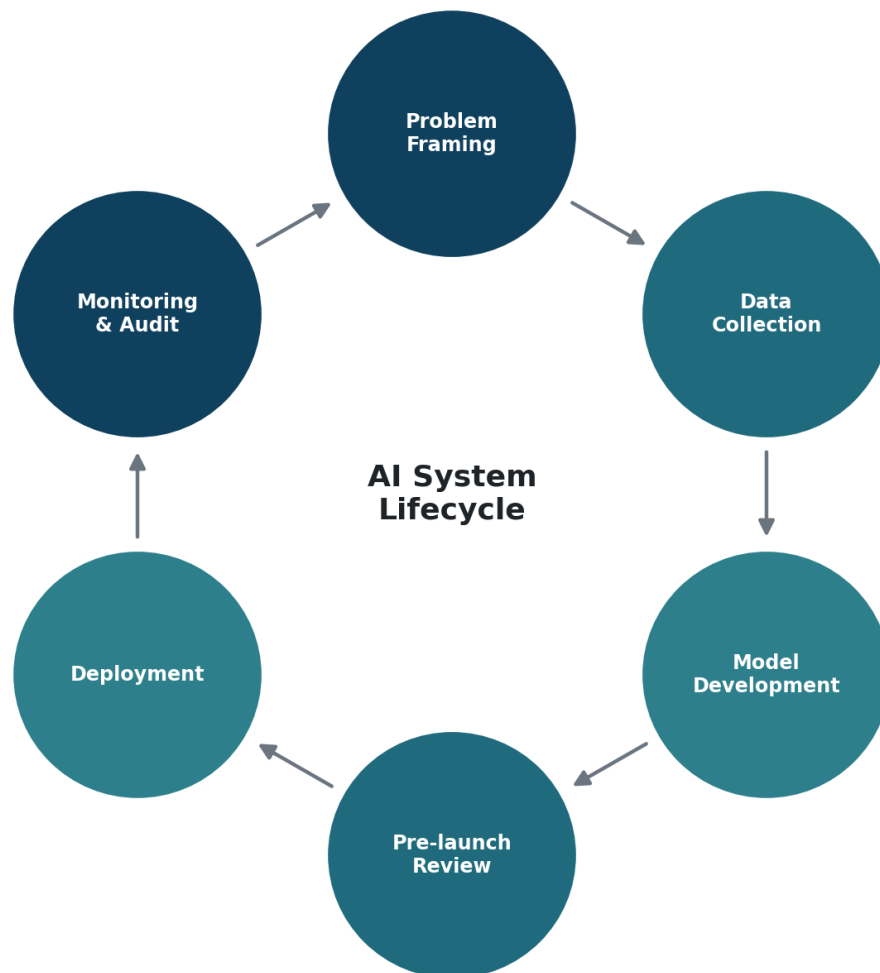


Figure 1.1 — The AI system lifecycle referenced throughout this course.

- **Module 2** establishes why ethical reasoning is inseparable from technical design choices, using Amazon's abandoned AI recruiting tool as a worked example.
- **Module 3** examines documented patterns of AI failure through the 2020 UK exam grading algorithm and the five categories of AI risk.
- **Module 4** maps the global AI regulatory landscape — the EU, UK, US and Gulf states.
- **Module 5** introduces governance structures using the NIST AI Risk Management Framework and the Dutch childcare benefits scandal as a governance failure case.
- **Module 6** covers the UK GDPR and Data Protection Act 2018 in detail, including automated decision-making and Data Protection Impact Assessments.
- **Module 7** translates governance into implementation across the delivery pipeline.

- **Module 8** governs generative AI, large language models and AI agents specifically, using the Samsung and Air Canada cases.
- **Module 9** applies the whole course to the data centre, including social media and media conduct.
- **Module 10** consolidates the course ahead of the Final Certification Exam.

1.3 Who This Course Is For

You do not need a background in philosophy or law to complete this course. You do need direct or indirect involvement with AI systems — building them, deploying them, procuring them, or being accountable for their outcomes. The frameworks presented here are intentionally cross-functional, because responsible AI is rarely the product of a single role or department: data centre operations and facilities staff, security and access control personnel, engineering and product teams, risk and compliance functions, recruitment and HR professionals placing people into AI-adjacent roles, and executives and non-executive directors all have a stake in getting this right, and each is addressed directly somewhere in this course.

This edition of the course pays particular attention to the physical infrastructure that makes AI possible: data centres. Whether you are a facilities engineer, a security officer, a network operations technician, a site manager, or a recruitment professional placing people into these roles, the systems described in this course are not abstract — they run on racks, in halls, and behind access-controlled doors that you or your colleagues manage every day. Module 9 is dedicated entirely to that operational context.

A Working Definition

For the purposes of this course, **AI ethics** refers to the principles and reasoning used to evaluate whether an AI system's design and behaviour are justifiable given its effects on people. **AI governance** refers to the organisational structures, policies and processes that turn those principles into consistent, accountable practice. Ethics asks what is right; governance asks how an organisation ensures it consistently does it.

IN PRACTICE

A mid-sized recruitment company realises it already uses three AI tools — a CV screening plug-in, a chatbot for candidate queries, and a scheduling assistant — none of which anyone formally approved. Its first governance step is not a new policy document; it is a one-page inventory (Appendix, Template 3) listing what each tool does, who owns it, and what data it touches. That single exercise is usually enough to reveal which systems need a closer look first.

1.4 The UK Policy Landscape

The United Kingdom has taken a deliberately different regulatory path from the European Union. Rather than a single horizontal AI statute, the UK Government's approach — set out by the Department for Science, Innovation and Technology (DSIT) — relies on existing regulators (such as the ICO, the Financial Conduct Authority, the Equality and Human Rights Commission and Ofcom) applying five cross-sectoral principles to

AI within their existing remits: safety and security, transparency and explainability, fairness, accountability and governance, and contestability and redress. This course is structured around those same underlying ideas, alongside the UK GDPR obligations covered in Module 6.

UK IN FOCUS

Throughout this course, "UK GDPR" refers to the retained version of the EU General Data Protection Regulation as it forms part of UK law, read together with the Data Protection Act 2018 and, from 2025 onwards, amendments introduced by the Data (Use and Access) Act 2025. Module 6 covers this in full.

CASE STUDY — OFQUAL (A PREVIEW)

In August 2020, England's exams regulator, Ofqual, used a statistical algorithm to assign A-level grades after COVID-19 cancelled exams. The algorithm downgraded roughly 40% of teacher-predicted grades, and disadvantaged students at larger state schools were hit hardest, while students at small, often fee-paying schools were more likely to be marked up. Within days, the Government reversed course, Ofqual's chief regulator resigned, and the episode remains one of the clearest public examples of an AI-adjacent system failing on fairness, transparency and accountability all at once. Module 3 examines exactly what went wrong and why.

1.5 Why Data Centres Sit at the Centre of This Course

AI systems do not run in the abstract. Every model referenced in this course — whether it is screening a CV, pricing an insurance policy, or triaging a patient — is trained and served from physical infrastructure, and an increasing share of that infrastructure sits in the United Kingdom. The UK has one of the largest data centre markets in Europe, concentrated around London, Slough and the wider “Western Corridor”, with continued growth driven specifically by demand for AI training and inference capacity. The Government has designated data centres as Critical National Infrastructure, placing them in the same protective category as energy and water systems.

That status has practical consequences for anyone working in or around these sites. It raises the security bar, increases regulatory and media attention, and means that decisions made in Modules 2 to 9 of this course — about fairness, failure, governance and data protection — are frequently implemented, monitored or breached at the physical layer this sector operates. Module 9 returns to this in detail, including what to do if a journalist or a piece of your own social media activity brings that physical layer into public view.

KEY TAKEAWAYS

- AI ethics and AI governance are related but distinct: ethics defines the standard, governance operationalises it.
- This course is built for practitioners across roles, not only for specialists in law or policy, and gives particular attention to data centre operations, security and facilities roles.
- The UK takes a principles-based, regulator-led approach to AI rather than a single AI statute, which this course reflects throughout.
- UK data centres are designated Critical National Infrastructure, which raises the security, regulatory and public scrutiny bar for everyone working on site.
- This course is built around real, documented cases — including the 2020 Ofqual grading algorithm and the Amazon and Dutch cases ahead — not hypothetical scenarios.

FURTHER READING

- gov.uk — A pro-innovation approach to AI regulation (DSIT white paper)
- ICO — guidance on AI and data protection, ico.org.uk
- OECD.AI Policy Observatory — oecd.ai

MODULE 1 CHECK-IN QUIZ

Five questions to confirm your understanding before moving on. Answers are provided in the Answer Key at the end of the course.

1. What best distinguishes AI governance from AI ethics, as defined in this course?

- A. Governance is optional; ethics is mandatory.
- B. Ethics defines the standard for right conduct; governance operationalises it consistently.
- C. They are interchangeable terms for the same activity.
- D. Governance applies only to public sector AI.

2. Which body leads the UK Government's overall policy approach to AI regulation?

- A. The Information Commissioner's Office (ICO) alone
- B. The Department for Science, Innovation and Technology (DSIT)
- C. The European Commission
- D. The Bank of England

3. The UK's approach to AI regulation, as described in Module 1, is best characterised as:

- A. A single comprehensive AI statute
- B. No regulation of AI at all
- C. A principles-based approach applied by existing sector regulators
- D. Regulation only for generative AI

4. In the hiring-platform framing example, which question is a governance question rather than an ethics question?

- A. Are the model's screening criteria fair and justifiable?
- B. Who approved the model for use and how are its outcomes monitored?

- C. What philosophical theory of fairness applies?
- D. Is the model technically accurate?

5. According to Module 1, why is this course designed for practitioners across multiple roles?

- A. Because only lawyers are responsible for AI ethics
- B. Because responsible AI is rarely the product of a single role or department
- C. Because engineers do not need to understand governance
- D. Because it is required by the Equality Act 2010